Analyzing the Spatio-Temporal Pattern of Disturbances in Yuzhong Mining Areas Using Continuous Variation Detection and Intensive Time-Series Landsat Observation

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***Abstract*—Disturbance detection and restoration monitoring are essential in mining regions for understanding landscape dynamics and supporting restoration decision-making. However, traditional remote sensing approaches often suffer from sparse temporal observations and phenological noise, which limit their capacity to capture subtle or short-term ecological dynamics. To overcome these issues, we develop an enhanced intensive time series LandTrendr (ITSLT) framework based on dense Landsat time series derived from all available surface reflectance imagery from 2000 to 2023, and integrate key temporal breakpoints for improved disturbance extraction. In addition, a dynamic time warping (DTW)-based trajectory similarity analysis is introduced to classify disturbance–recovery patterns and quantify deviation from reference baseline sequences. Applied to the Yuzhong mining area in central Henan, China, the proposed method improves the temporal sensitivity and robustness of disturbance characterization. Results show pronounced spatial heterogeneity in disturbance magnitude and recovery trajectories. Human-induced disturbances account for 52.91% of all events while mining-related impacts contribute 2.61% and 5.98% of the total area, highlighting the dominant role of anthropogenic activities. Although regional ecosystem conditions exhibit an overall recovery trend, they have not yet returned to the levels observed in 2000. Disturbances have progressively expanded from core mining zones toward adjacent construction and transportation corridors, reflecting a shift from localized to more spatially connected anthropogenic impacts. Our combined ITSLT with DTW framework provides an effective tool for long-term disturbance detection and ecological recovery assessment in mining landscapes, and supports application in ecological restoration monitoring and green mine planning.**

***Index Terms*—Disturbance detection, Google Earth Engine (GEE), LandTrendr, mining areas, time-series analysis.**

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I. Introduction

MINING, as a pivotal industrial activity supporting global resource supplies, plays a significant role in driving economic growth and social development. At the same time, mining activities reshape surface landscapes and disrupt ecological processes at various spatial and temporal scales [1]. In recent years, increasing global attention has been devoted to the cascading ecological and environmental consequences associated with mineral resource extraction [2]. For instance, during the 28th Conference of the Parties to the United Nations Framework Convention on Climate Change, it was emphasized that the continued expansion of mining areas imposes a cumulative stress on natural systems, accelerates biodiversity loss, exacerbates regional water scarcity, and poses risks to human health through land fragmentation and environmental pollution [3]. As one of the most intensive and persistent forms of anthropogenic disturbance, mining introduces substantial constraints on regional sustainable development [4]. Consequently, the identification, assessment, and restoration of mining-induced disturbances have become critical research topics. Extensive studies have investigated disturbance responses and recovery mechanisms across diverse mining environments, including coal [5], granite [6], and rare earth mines [7], all of which experience direct or indirect impacts from extraction activities. In response to the challenges posed by high-intensity disturbance and ecological degradation, China's "13th Five-Year Plan" national key Research and development program listed mine ecological restoration as a priority, emphasizing the urgency and necessity of managing mining-induced environmental change. Against this background, accurately identifying different types of mining disturbance processes, elucidating their spatio-temporal evolution, and conducting fine-scale monitoring of postdisturbance recovery have emerged as key scientific challenges for mine ecosystem management.

Traditional methods for assessing mining disturbance and restoration, primarily dependent on field surveys and manual monitoring, suffer from long implementation cycles, high costs, and limited spatial coverage, making continuous and objective assessment of temporal ecosystem changes challenging [8]. Rapid advances in aerospace technologies, computing,

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and sensor design have considerably improved the spectral, spatial, and temporal resolution of remotely sensed imagery, providing powerful capabilities for accurately detecting disturbance processes and restoration dynamics [9]. These developments have facilitated the widespread application of remote sensing in geoscience and ecological research [10]. Typical applications include mining area delineation [11], land use/cover change monitoring [12], forest disturbance detection [13], and geological and lithological characterization [14]. Among these, remote sensing plays a particularly important role in mining environments, where it provides critical observational support for detecting disturbance processes and assessing ecological impacts [15]. For mining disturbance studies, the broad spatial coverage and high temporal continuity of remote sensing data enable detailed characterization of dynamic land-use transitions, vegetation degradation, and ecological recovery, significantly enhancing the precision and timeliness of monitoring [16]. Accordingly, remote sensing has become a core data source for long-term disturbance monitoring, postmining ecological restoration assessment, and evidence-based decision-making in ecological governance.

The Landsat satellite series, with more than four decades of uninterrupted global imagery, remains one of the most important data sources for monitoring long-term surface changes [17]. Early studies primarily relied on bitemporal or limited multitemporal observations to detect spatial differences between discrete time points or to fit simple temporal curves describing general trajectories of ecosystem conditions [18]. However, these approaches are constrained by sparse temporal resolution and the rapid pace of mining disturbances, often failing to capture fine-scale dynamics of extraction processes [19]. Although recent advances in remote sensing indices and ecological evaluation models have enabled periodic disturbance assessments of mining areas using imagery at 2-, 5-, or 10-year intervals, such intervals remain insufficient for identifying rapid or transient changes, thereby limiting the precision of fine-grained disturbance characterization [20].

Continuous time-series change detection models offer new opportunities to overcome these limitations. Trajectory-based change detection algorithms [21] can extract long-term disturbance trends in forests, while linear trend regression methods describe overall extraction dynamics. Continuous change detection and classification (CCDC) algorithms [22] have also been applied to identify long-term mining-induced changes. However, many existing approaches are constrained by sparse temporal observations, which limit their ability to capture rapid or short-term disturbance signals in highly dynamic environments [23], [24], [25], [26]. In addition, these methods often require complete, high-quality time-series inputs and rely on simplified assumptions about underlying temporal patterns. In mining areas characterized by frequent disturbances, such approaches may fail to capture fine-scale features and complex mechanisms, thereby reducing the accuracy and robustness of practical applications [27]. These limitations indicate that existing approaches are still insufficient to fully capture the complexity of mining-induced ecological dynamics. In particular, constraints in temporal representation and analytical integration hinder the accurate detection of subtle disturbances and the characterization of disturbance–recovery processes. Consequently, the spatio-temporal evolution of mining disturbances and their ecological impacts remains inadequately resolved.

To improve the detection and interpretation of mining disturbances, this study proposes an enhanced intensive time series LandTrendr (ITSLT) framework that utilizes all available Landsat imagery to construct a dense time series dataset. The method integrates key temporal change points to strengthen disturbance sensitivity and reduce omission errors. Furthermore, a dynamic time warping (DTW) similarity metric is incorporated to classify disturbance–recovery trajectories and quantify deviations from reference sequences. Our method is evaluated in a study area located in Henan Province, China, a major region for nonferrous and precious metal mining, where decades of intensive extraction have significantly deteriorated regional ecological quality. In particular, the Yuzhong region (Zhengzhou–Pingdingshan–Xuchang corridor) exhibits concentrated mining activities and diverse disturbance types, making it a representative area for evaluating mining-induced ecological change. There is an urgent need to monitor surface disturbances, explore the drivers of ecological environment quality dynamics, strengthen environmental governance, and support sustainable economic, social, and ecological development.

The main contributions of this article can be summarized as follows.

1) An improved ITSLT framework is proposed to integrate and optimize mining disturbance breakpoints, ensuring accurate detection and retention of disturbance information.
2) The framework is combined with the DTW algorithm to classify and distinguish multiple types of mining disturbances in Henan province, enabling robust characterization of complex disturbance patterns.
3) The enhanced ITSLT model is applied to analyze the spatio-temporal evolution of mining disturbances, providing scientific evidence for understanding regional ecological degradation mechanisms and supporting targeted ecological restoration strategies.

II. Materials and Methods

A. Study Area

The study area is situated in the southern portion of the middle and lower reaches of the Yellow River in China, encompassing the administrative regions of Zhengzhou, Pingdingshan, and Xuchang, with a total area of approximately 20 445 km^2 (Fig. 1) [28]. The region experiences a warm temperate continental monsoon climate, characterized by distinct seasonal variation and concentrated summer precipitation [29]. Its topography comprises a heterogeneous assemblage of plains, hills, and low mountain terrain [30]. As a nationally significant polymetallic resource-rich region, the area contains China's second-largest proven bauxite reserves, while Pingdingshan alone holds approximately nine billion tons of coal resources [31]. However, decades of extensive mechanized mining have resulted in substantial ecological and

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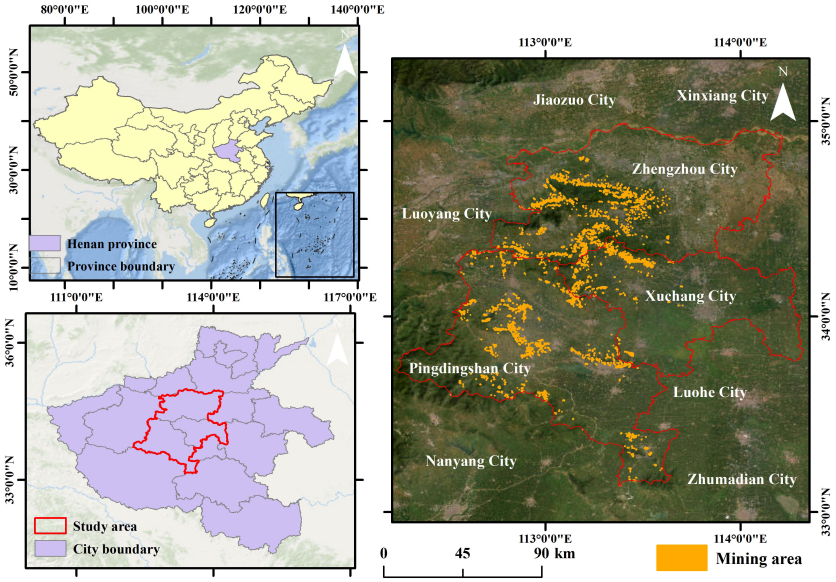


Fig. 1. Location of the study area.

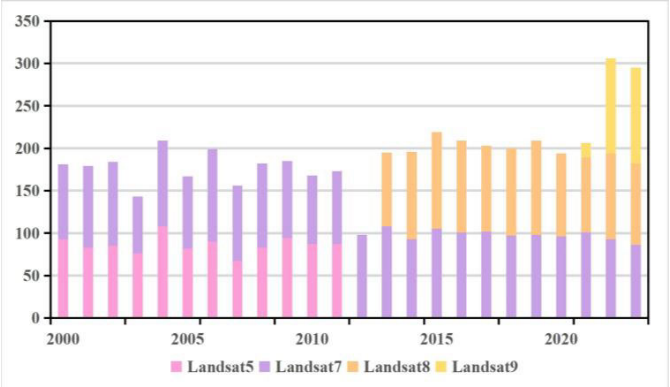


Fig. 2. Statistical chart of the four types of Landsat data used in the study area.

environmental challenges [32]. Unregulated extraction has led to the uncontrolled discharge of waste rock and wastewater, triggering geological hazards such as ground subsidence, landslides, and debris flows, thereby exerting severe impacts on regional ecosystems [33]. According to the "Henan Province Mineral Resources Master Plan (2016–2020)," historically abandoned mines requiring remediation cover an estimated 27 800 hectares, with accumulated tailings and solid waste reaching 867.28 million tons [34].

To mitigate these impacts, the region has actively implemented mine geological environment restoration and comprehensive management since the 1990's, carrying out ecological reclamation and urban greening projects. These efforts have contributed to national recognitions such as "National Forest City" and "National Garden City," reflecting preliminary success in transitioning mining cities toward ecological sustainability [35].

B. Datasets

This study utilized a total of 4656 surface reflectance images acquired from Landsat TM, ETM+, and OLI sensors for the period 2000–2023, obtained via the Google Earth Engine (GEE) platform. These images cover all the cloud-free data collected throughout the year, thus ensuring a dense time coverage each year. The image data used are shown in Fig. 2 and Table I. All images underwent atmospheric correction and standard preprocessing in GEE using the LEDAPS and LaSRC algorithms, while pixels affected by clouds, shadows, snow, or ice were masked using optical bands information and the corresponding quality assessment bands [36]. To mitigate radiometric discrepancies arising from differences in spectral response functions among the three Landsat sensors, a statistical harmonization function proposed by Roy et al. [37] was applied, ensuring spectral consistency across the multisensor time series. Annual composite images were subsequently generated using median surface reflectance, producing one composite image for each year. After completing these preprocessing steps, we constructed a pixel-level spectral index time series dataset, resulting in 24 annual composite images covering the entire study period.

C. Overview of the Proposed Methodology

The overall methodological framework of this study is illustrated in Fig. 3 and consists of four major components: (a) data preparation, (b) construction of intensive time-series with LandTrendr, (c) classification of disturbance trajectories, and (d) assessing gradual change. First, Landsat surface reflectance imagery was preprocessed to generate a continuous 2000–2023 time series dataset, from which the normalized difference vegetation index (NDVI) was derived. Second, considering the year-round nature of mining activities and the complexity of ecological change processes, a dense time-series disturbance detection model was developed by integrating a moving-window strategy with the LandTrendr algorithm. Third, the DTW method was employed to compute the distance between disturbance and reference baseline

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TABLE I
DATASETS USED IN THIS STUDY

Data	Description	Data source
Landsat TM/ETM+/OLI	4655 images of Surface Reflectance Tier1 of 2000–2023	Surface Reflectance Tier1 dataset available from GEE
Reference data	The distribution information of mining area on the official website is used as reference data.	https://www.henan.gov.cn
Digital Elevation Model (DEM)	DEM with 30-m spatial resolution	http://www.gscloud.cn/
Landsat TM/ETM+/OLI	4655 images of Surface Reflectance Tier1 of 2000–2023	Surface Reflectance Tier1 dataset available from GEE

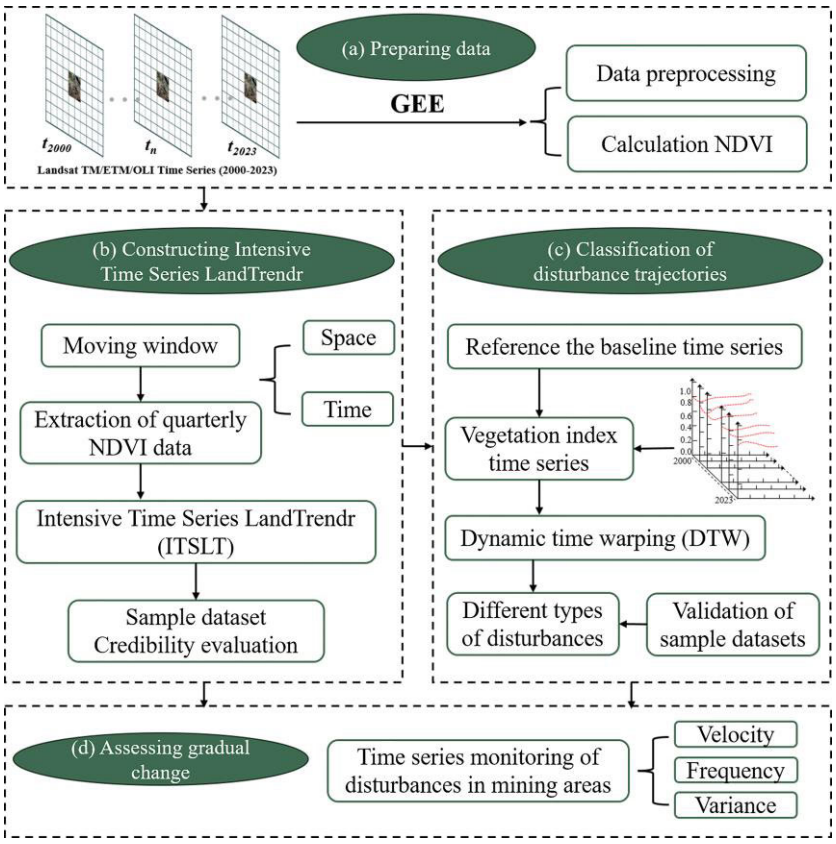


Fig. 3. Workflow of the proposed methodology. (a) Data preparation, (b) construction of intensive time-series with LandTrendr, (c) classification of disturbance trajectories, and (d) assessing gradual change.

sequences, enabling the classification of disturbance types and characterization of their evolution. Finally, the spatio-temporal patterns of mining disturbances were comprehensively evaluated from multiple perspectives to obtain a detailed understanding of disturbance dynamics.

D. Data Preparation

Previous studies have demonstrated that NDVI is an effective indicator for detecting vegetation changes induced by mining activities [38]. In this study, NDVI was employed to characterize vegetation growth conditions. Given that vegetation pixels exhibit distinctive phenological patterns at the annual scale, their peak NDVI values provide a reliable representation of vegetation cover variation and can effectively capture mining disturbances [39]. It remains a robust and informative metric for monitoring vegetation dynamics. Accordingly, the annual maximum NDVI was adopted to represent yearly vegetation growth, resulting in a 24-year NDVI dataset spanning 2000–2023. The calculation is expressed as follows:

$$NDVI = \frac{\rho_{NIR} - \rho_{Red}}{\rho_{NIR} + \rho_{Red}} \tag{1}$$

where ρ_{NIR} and ρ_{Red} represent the surface reflectance values in the red and near-infrared bands of Landsat imagery, respectively.

E. Construction of ITSLT

1) LandTrendr Algorithm: The LandTrendr algorithm extracts per-pixel trajectories of spectral bands or indices,

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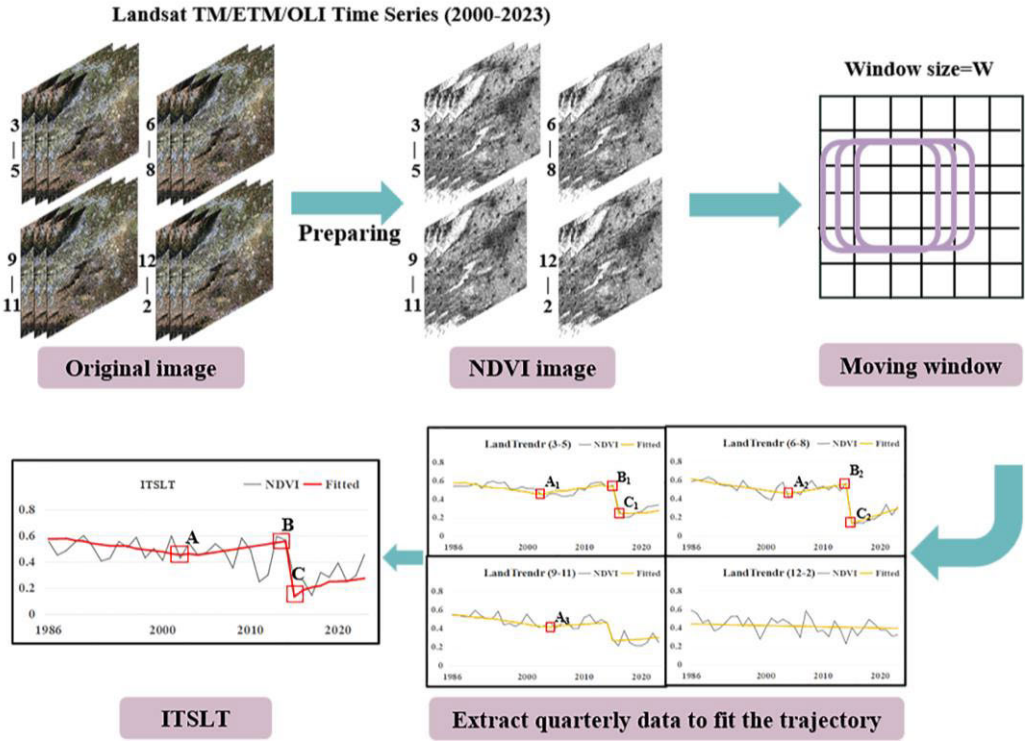


Fig. 4. ITSLT algorithm fits trajectories using all available Landsat image data.

effectively suppressing noise while preserving key vertex information [40]. The identified vertices segment the trajectory into intervals of persistent change or stability, recording both the timing and magnitude of transitions and forming a set of spectral line segments defined by time and spectral value [41]. Using the temporal and spectral attributes at segment endpoints, the duration and magnitude of changes can be calculated, enabling the extraction of disturbance characteristics such as timing, frequency, duration, and magnitude, along with recovery metrics and predisturbance baselines [42].

2) Intensive Time Series LandTrendr: To minimize noise introduced by phenological variability and changes in solar elevation angle, the LandTrendr algorithm typically relies on imagery acquired on similar dates each year for disturbance detection [43]. However, the limited number of such images results in relatively sparse temporal sampling, which restricts the ability to capture the small-amplitude phenological patterns and short-term ecological fluctuations that are common in mining environments [44]. Therefore, incorporating all available Landsat observations to form a dense time series is necessary to enhance temporal resolution, enrich information content, and improve the sensitivity of mining disturbance detection.

To address this limitation, all available Landsat observations were incorporated to construct a dense intraannual time series. In particular, cloud-free observations within each year were grouped into quarterly intervals (e.g., 3–5, 6–8, 9–11, and 12–2), and median compositing was applied to generate representative images for each period, thereby reducing noise and ensuring temporal consistency. This strategy increases temporal resolution while preserving key disturbance signals.

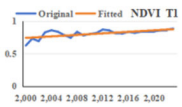
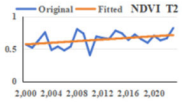
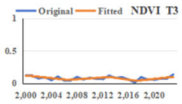
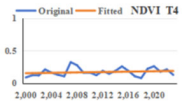
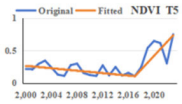
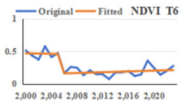
A key advantage of LandTrendr lies in its ability to simultaneously capture long-term gradual trends and short-term abrupt disturbances through adaptive temporal segmentation. By tracking multiyear vegetation dynamics in mining areas, the algorithm models vegetation degradation and recovery processes with high temporal robustness [45]. This study takes NDVI as the input index and calculates the NDVI distribution value within the sliding window item by item by moving the spatial window to the position of adjacent pixels. It detects the trend of vegetation destruction and recovery during the disturbance stage in the mining area, obtains the evolution process of the original vegetation state at the pixel level, and then obtains the disturbance change trajectory of the mining area in the study area. Based on this, ITSLT detects the time series curves generated by the same quarter data throughout the monitoring period, and combines the key time change points in the detection results: if a key change point occurs only once, it is directly recorded; if it occurs multiple times, the event with the largest disturbance level is retained (Fig. 4). This integration strategy enables ITSLT to avoid missed detections of subtle disturbances present in the original LandTrendr results and allows earlier identification of disturbance onset, thereby reducing false detections. At the same time, the method enhances the responsiveness and variability of spectral time-series trajectories, substantially improving the spatio-temporal precision of mining disturbance detection.

F. Classification of Disturbance Trajectories

1) Algorithm DTW: DTW is a classical method for measuring similarity between time series by identifying an optimal

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TABLE II
CLASSIFICATION OF DIFFERENT DISTURBANCE TYPES IN MINING AREAS

Type	Description	Image examples	Typical trajectory	
Non-disturbances	During the monitoring period, the natural vegetation cover is maintained and NDVI values tend to be stable.			T1
	During the monitoring period, the land cover is cultivated land, and NDVI values tend to be stable.			T2
	During the monitoring period, the ground cover is buildings, and the NDVI curve is basically stable.			T3
Mining disturbances	During the monitoring period, it is always mining areas, generally the core area of mining, and there is always mining disturbance or poor recovery after stopping mining, and NDVI value tends to be stable.			T4
	The NDVI curve shows an upward trend.			T5
Building disturbances	During the monitoring period, there are construction activities, which cause interference. The NDVI curve shows a downward trend.			T6

alignment that minimizes the cumulative distance between two sequences [46]. In this study, DTW was applied to quantify the deviation between each pixel-level NDVI trajectory and its corresponding reference baseline sequence [47].

To ensure comparability, all NDVI time series were normalized prior to analysis. Thereby accommodating potential temporal shifts in vegetation response caused by disturbance and recovery processes. The resulting cumulative distance along the optimal alignment path is defined as the DTW distance, which serves as an indicator of trajectory deviation. Larger DTW values indicate greater divergence from expected vegetation dynamics and are therefore associated with stronger disturbance signals.

2) Classification of Disturbance Types in Different Mining Areas: Based on visual interpretation of Google Earth imagery, vegetation, cropland, and built-up areas were identified as the dominant land-cover types surrounding mining zones. These classes exhibit distinct NDVI characteristics, enabling effective discrimination. To refine disturbance identification, LandTrendr thresholds were adjusted using the interactive module, and ITSLT was applied to analyze trajectory patterns associated with specific disturbance types (Table II).

Subsequently, the DTW algorithm was used to construct NDVI reference baseline time series for each disturbance type in the mining region. Each baseline was generated via double-logistic curve fitting, with parameters estimated through a least-squares approach based on selected training pixels to reduce interannual seasonal variation. The DTW distance between each NDVI time series and its baseline reflects the degree of deviation. Under disturbance conditions, the NDVI trajectory exhibits notable shifts or abrupt declines, resulting in increased DTW distance. Under stable conditions, DTW distances remain within a defined range; deviations beyond this threshold indicate disturbance. This framework enabled classification of disturbance–recovery types and systematic evaluation of temporal dynamics within a 5-km buffer zone surrounding the mining areas. This study ensures adaptability to regional characteristics of mining disturbance, thereby enhancing the robustness and transferability of the proposed framework.

G. Validation

Accuracy assessment was conducted through visual interpretation using Google Earth ground-change information and historical high-resolution imagery, a widely used approach for evaluating classification reliability in surface-change detection studies [48]. Mining disturbances were identified within the period 2001–2023. Accordingly, 500 random points were selected each year within the disturbed area, resulting in a total of 11 500 validation samples. Each sample point was manually labeled through interactive interpretation of high-resolution Google Earth imagery to determine its corresponding disturbance year. In this study, the accuracy of the affected years was monitored, and the years detected as interference by the algorithm were compared with the reference years to evaluate the temporal consistency.

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TABLE III
DETECTION ACCURACY OF DISTURBANCE YEARS IN THE CONSIDERED MINING AREA

Years	F1 Score	Years	F1 Score	Years	F1 Score	Years	F1 Score
2000	0.89	2006	0.91	2012	0.90	2018	0.88
2001	0.81	2007	0.81	2013	0.93	2019	0.91
2002	0.92	2008	0.89	2014	0.92	2020	0.87
2003	0.88	2009	0.77	2015	0.91	2021	0.88
2004	0.90	2010	0.79	2016	0.86	2022	0.83
2005	0.90	2011	0.91	2017	0.88	2023	0.94

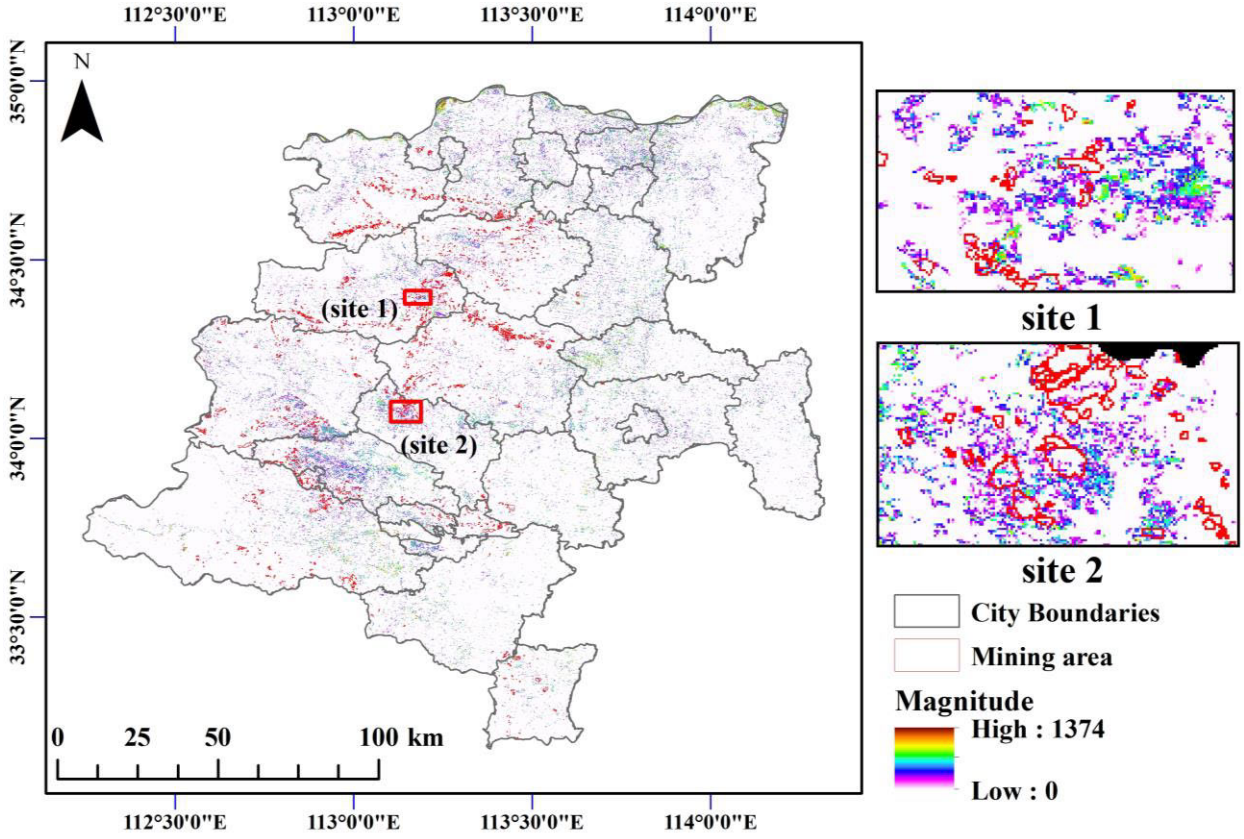


Fig. 5. Spatial distribution map of maximum disturbance magnitude in the study area and detailed view of two mining areas.

Based on these validation samples, a confusion matrix was generated for each study unit, and the producer accuracy (PA), user accuracy (UA), overall accuracy (OA), Kappa coefficient (Kappa), and $F1$ score were calculated [49]. We also calculated the average harmonic value $F1$, which can reconcile the average of UA and PA to more effectively characterize the accuracy of verification points in different years. Its value ranges from 0 to 1, with higher values indicating better classification performance.

III. RESULTS

A. Accuracy Assessment

The OA of the disturbance detection results was 86.65%, with a Kappa coefficient of 0.86. The $F1$ score ranged from 0.77 to 0.94 across different disturbance years (Table III). Although the OA value was high, the $F1$ scores exhibited notable variability. In certain years, the accuracy of identifying year disturbance was relatively low; for instance, the $F1$ scores for reclamation in 2009 and 2010 were both below 0.8, likely due to the lower quality of remote sensing data available during those years.

B. Mining Disturbance Detection

1) Spatial Variation Analysis of Mining Disturbances: Based on the ITSLT algorithm, the maximum disturbance magnitude to the disturbance in the mining area was visually displayed, with mine area boundaries delineated in red. The maximum disturbance magnitude represents the largest change in NDVI detected for each pixel during the monitoring period, indicating the intensity of vegetation disturbance. As shown in Fig. 5, zones of high disturbance intensity were primarily concentrated within core mining areas and adjacent major urban centers, including Zhengzhou in the north, Xuchang in the east, and Pingdingshan in the south. Fig. 5 (site 1) and (site 2) provide detailed views of the Yuzhou coal mines and northern Pingdingshan mining sites, clearly revealing sub-

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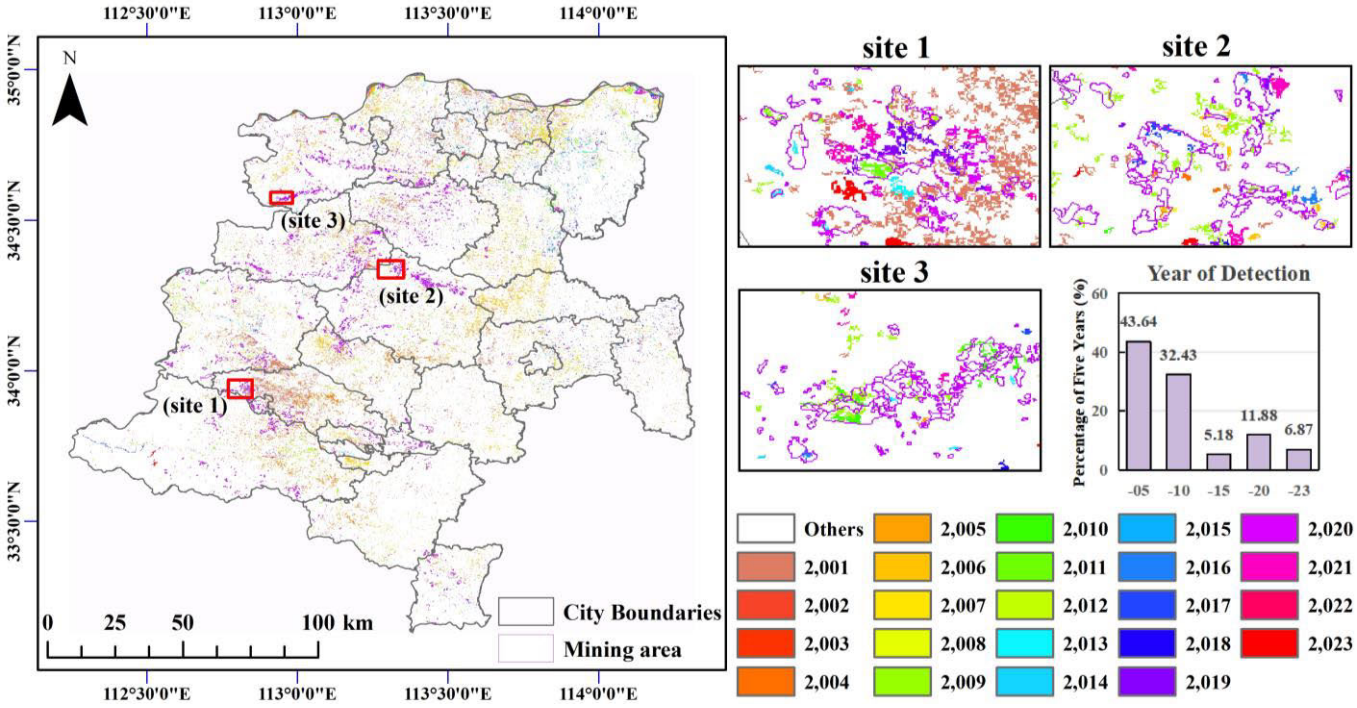


Fig. 6. Disturbance year profiles and details of three mining areas extracted from the considered study area.

stantial mining disturbances caused by long-term resource extraction, stripping operations, waste-rock accumulation, and phased reclamation. These areas commonly exhibited fragmented land cover, reduced vegetation continuity, and intensified human activities, contributing to elevated disturbance magnitudes [50].

On the contrary, the mountains in the southwest and center of the study area were predominantly covered by forested terrain, maintaining relatively intact ecological structures and exhibiting strong self-regulation capacities. Vegetation in these regions demonstrated higher resilience to external disturbances, supporting a gradual decline in forest disturbances and overall stability throughout the study period [51].

At the regional scale, disturbance patterns displayed notable spatial heterogeneity directly aligned with underlying disturbance sources. Disturbances near mining sites were highly concentrated due to intensive mechanized extraction and severe land fragmentation. In the plains, disturbances were primarily driven by urban expansion and industrial restructuring, where rapid increases in construction land enhanced spatial connectivity and intensified disturbance levels. Overall, the interaction among mining, urban, and natural systems produced a characteristic gradient of anthropogenic disturbance across the region, emphasizing the importance of long-term ecological monitoring and restoration efforts.

2) Temporal Analysis of Mining Disturbances: Using the ITSLT algorithm, mining disturbances in the mining area (from 2000 to 2023) were detected, with mine area boundaries delineated in purple, and the timing and spatial distribution of maximum-magnitude disturbances were illustrated in Fig. 6. Detailed views of three representative mining sites (Sites 1–3) were provided in Fig. 6 (site 1)–(site 3). Statistical analysis indicated that during the early years of the study period, mining activities in the mining area were extensive and poorly regulated. Waste rock and wastewater were frequently discharged directly during extraction and processing, leading to secondary geological hazards such as ground subsidence, landslides, and debris flows [52]. These hazards not only caused direct surface damage but also accelerated vegetation loss, disrupted soil structure, and altered hydrological pathways, resulting in widespread ecosystem degradation. Consequently, the highest disturbance magnitudes were predominantly observed before 2010.

Following the implementation of land reclamation policies and the mine geological environment restoration programs, disturbances attributable to mining activities were partially reduced in some areas. On the contrary, disturbances associated with urban expansion were primarily concentrated around major city centers, with both their temporal and spatial patterns closely aligned with periods of rapid urban development. The expansion of construction land modified surface cover, increased land fragmentation, and imposed additional thermal and hydrological pressures, resulting in sustained ecological impacts [53].

In mountainous regions, long-term disturbances persisted due to management challenges and limited environmental governance capacity, accounting for the continued presence of high-intensity disturbance zones [54]. Overall, the timing and magnitude of disturbances exhibit strong correspondence with extraction activities. However, due to complex surface conditions and the slow pace of ecological response, reclamation and ecological restoration often experience significant delays, requiring several years or even decades to achieve substantial recovery. This extended restoration trajectory highlights the importance of continuous disturbance monitoring and reveals the dual characteristics of vulnerability and resilience

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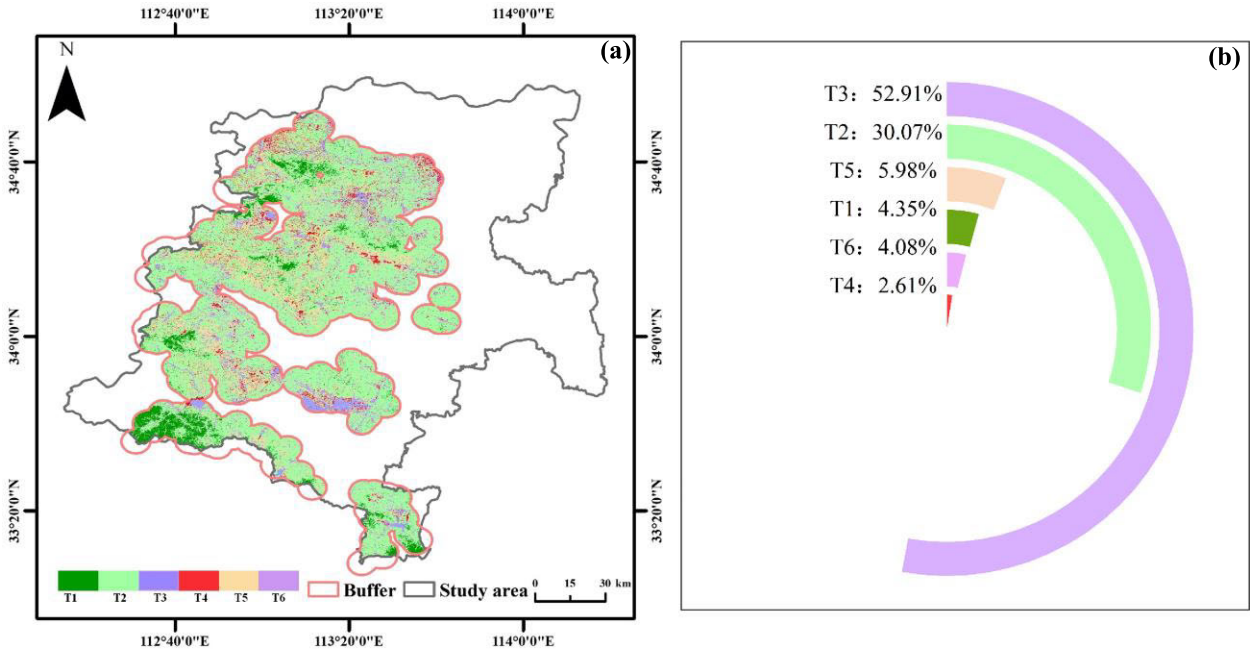


Fig. 7. Classification results of different bypass types in mining areas. (a) Classification result diagram and (b) statistical chart of the classification results.

exhibited by regional ecosystems under prolonged intensive anthropogenic pressures.

C. Changes in Ecological Disturbances in Mining Areas

The classification results were evaluated using a confusion matrix, yielding an OA value exceeding 85% and a Kappa greater than 0.80, meeting the requirements for subsequent analyses. Misclassifications occurred primarily in transitional or overlapping zones between disturbance types, reflecting inherent ecological complexity.

From the classification results in Fig. 7, T1 represents areas least affected by mining disturbances, where large, contiguous patches of natural vegetation remained stable throughout the monitoring period, particularly in the southwestern Funiu Mountains and Central Song Mountains. These regions, characterized by complex terrain and minimal human interference, maintained high ecological integrity [55].

T3, which is most strongly influenced by human activities, accounts for 52.91% of the study area. Disturbances in this category are primarily associated with reclamation, residential construction, and infrastructure development, reflecting sustained ecological pressure from urban expansion and regional development. T2 covers 30.07% of the area and is mainly related to agricultural activities. Although T2 and T3 exhibit relatively stable NDVI values, this stability is largely driven by continuous human management rather than natural ecological processes. As a result, these areas still experience significant ecological disturbance in terms of land-use transformation, habitat fragmentation, and reduced ecosystem resilience [56].

Mining-related disturbances (T4 and T5) account for 2.61% and 5.98% of the area, respectively. Especially before 2010, due to insufficient policy supervision and limited recovery measures, the number of pixels in the recovery stage was relatively low, and the overall condition showed a degradation trend. After 2010, strengthened mining management policies and the gradual implementation of ecological restoration initiatives significantly enhanced postmining reclamation. Consequently, both the number of recovering pixels and the magnitude of ecological improvement increased, particularly in partially rehabilitated open-pit mining areas, demonstrating the positive effects of restoration practices [57].

T6 represents disturbances associated with construction activities, where NDVI exhibits a clear downward trend due to vegetation removal and land surface alteration. These areas are mainly distributed around expanding urban zones and infrastructure corridors, reflecting short-term but intense anthropogenic disturbance. Compared with mining-related disturbances, T6 is typically characterized by more abrupt changes and limited natural recovery within the monitoring period.

D. Characteristics of Ecological Disturbance in Representative Mining Areas

The study area includes extensive mining sites as well as surrounding cropland, urban areas, and rural settlements. With the expansion of mining activities and the intensification of extraction methods, mining-induced ecological disturbances have become increasingly diverse in type, spatial extent, and process complexity. To illustrate long-term disturbance–recovery dynamics, two representative mining sites were selected based on the spatial distribution of identified disturbances.

Fig. 8(a) shows the open-pit bauxite mining areas in Jia County, Pingdingshan. This site represents a "mine–farmland–urban" composite disturbance zone, where mining areas are surrounded by cropland and residential settlements characterized by high levels of human activity. Disturbances are primarily driven by agricultural cultivation and the expansion of construction land. Despite limited postextraction restoration efforts, ecological degradation con-

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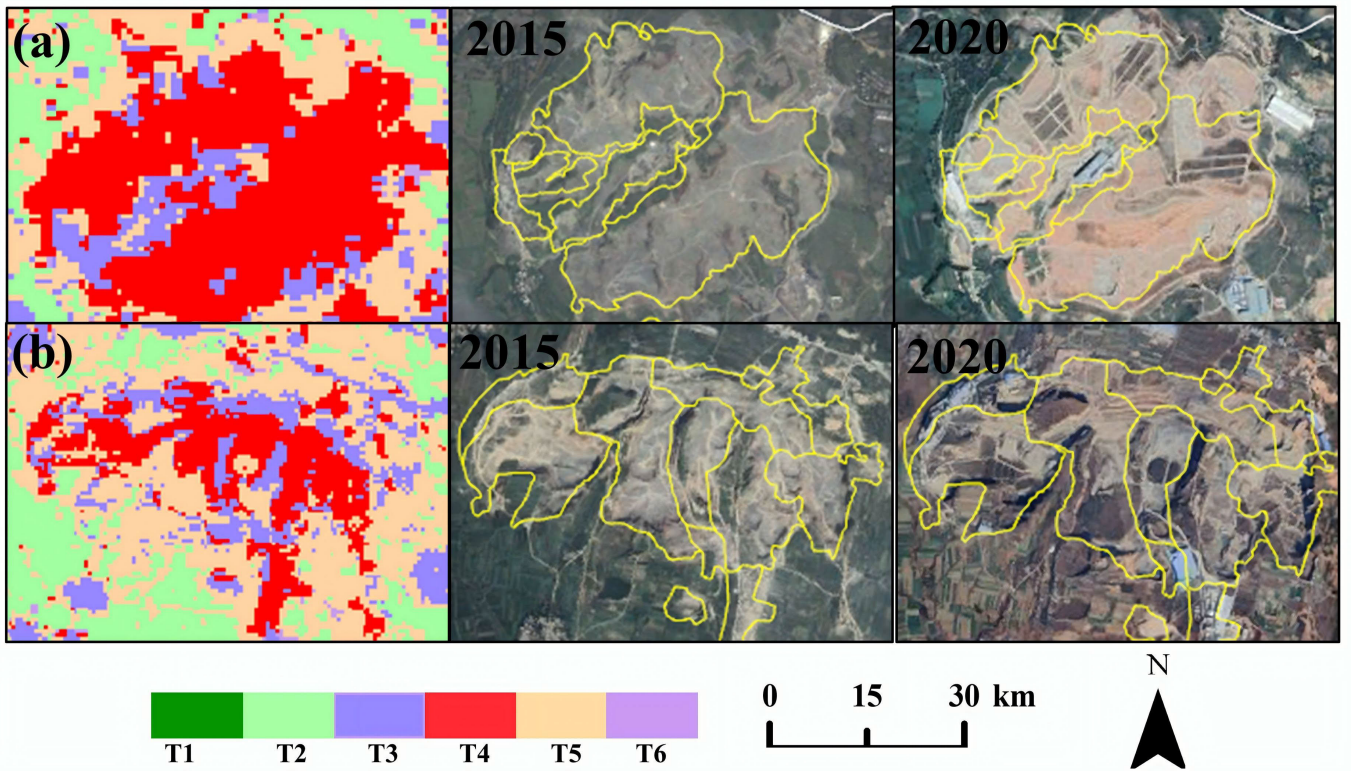


Fig. 8. Classification of typical mining areas. (a) shows the open-pit bauxite mining area in Jia County, Pingdingshan City, while (b) shows the open-pit limestone mining area in Wuliang Town, Yuzhou.

tinued to expand outward from the central mining area from 2000 to 2023, reflecting the imbalance between resource extraction and restoration activities as well as persistent anthropogenic pressures.

Fig. 8(b) illustrates the open-pit limestone mining area in Wuliang Town, Yuzhou. Mining activities at this site have ceased after operations peaked in the 1980's and 1990's. During that period, the growth of the township's collective economy, along with the proliferation of lime kilns and gravel plants, stimulated local development but caused severe ecological degradation. Outdated equipment and unregulated extraction accelerated the deterioration of mountain ecosystems, resulting in widespread environmental damage. To address these issues, the "Abandoned Mine Ecological Restoration Implementation Plan" was proposed and implemented. Guided by the principles of "forest where suitable, grass where suitable, landscape where suitable, construction where suitable," and tailored to local topography and soil conditions, the plan provided a systematic framework for restoration [58]. By the end of 2020, Wuliang Town had restored 32 abandoned hills, slopes, and mining pits, covering approximately 24 km^2, substantially improving regional ecological functions. Exposed rocky surfaces were mountains that gradually revegetated, forming ecological landscapes such as parks, gardens, and forests. At the same time, ongoing socio-economic development has introduced new construction-related disturbances in certain areas, underscoring the continued interaction between ecological restoration efforts and human activities.

IV. Discussion

A. Mining Disturbance Identification Process

Mining disturbances generally progress through four main stages: preextraction preparation, surface soil disturbance (primarily involving vegetation removal and topsoil stripping), intensive extraction, and subsequent ecological restoration [59]. Throughout these stages, mining activities cause rapid declines in vegetation cover, commonly observed as sharp decreases in indices such as NDVI. Continued interferences from ore crushing, stockpiling, and transportation maintain vegetation indices at low levels until either artificial reclamation is implemented or natural succession initiates vegetation recovery [60]. It should be noted that factors such as phenological cycles, interannual climate fluctuations, satellite data acquisition timing, and image quality can influence vegetation time series trajectories. However, mining-induced ecological degradation and subsequent recovery remain the primary drivers of vegetation dynamics in mining areas [60].

Building on this understanding, we developed an analytical framework to characterize temporal vegetation dynamics and support continuous monitoring of ecological conditions through pixel-level trajectory analysis. Fig. 9 presents representative vegetation trajectories. Using NDVI as the monitoring metric, our framework extracts disturbance and recovery processes from intrapixel time-series variations. Leveraging the ITSLT algorithm, NDVI effectively captures both abrupt disturbance events and subsequent recovery in open-pit mining areas, with all analyses implemented on the GEE platform.

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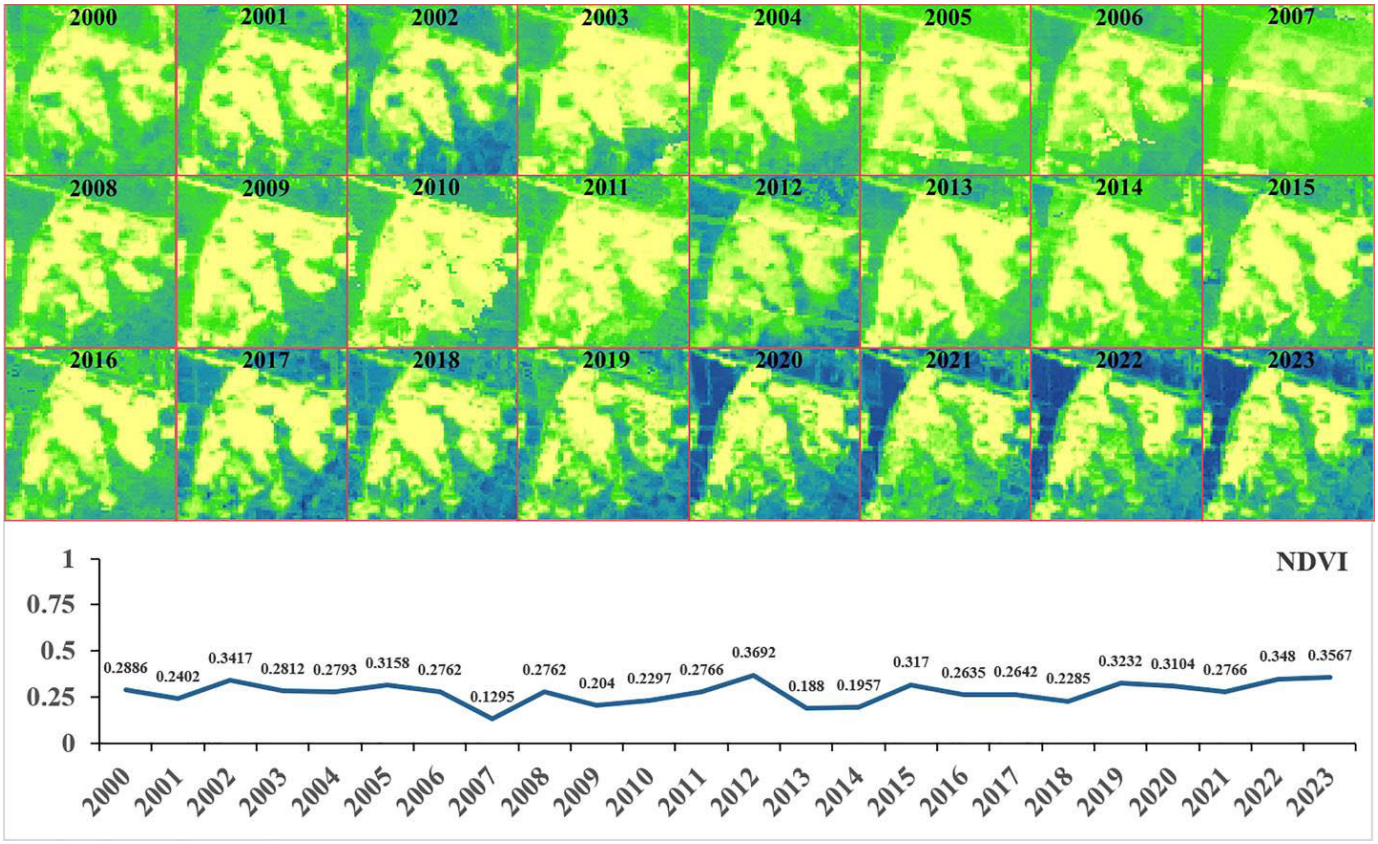


Fig. 9. Destruction and reclamation trajectories (at pixel scale) for mining-induced disturbance.

Compared with traditional time-series processing methods, GEE offers substantial advantages in large-scale data accessibility, cloud-computing efficiency, and integration of multisource datasets, thereby enabling more timely and accurate monitoring of mining disturbances [61]. The disturbance process itself inherently follows a nonlinear pattern characterized by sudden degradation followed by gradual recovery. Vegetation cover typically declines abruptly after extraction begins, whereas recovery progresses more slowly and remains sensitive to continued human activities, climatic fluctuations, and geological conditions [62]. Therefore, automated identification and classification of these stages not only elucidate the cascading ecological effects of mining activities but also provide a robust and reusable framework for mine environmental management, land reclamation planning, and ecological restoration assessment.

B. Recommendations for Postmining Ecological Restoration

Postmining ecological restoration represents a comprehensive natural resource management process rather than a set of isolated engineering interventions. Its primary objective is to restore ecosystem structural integrity and functional sustainability. Due to the extensive spatial disturbances and long-term ecological impacts associated with mining, restoration efforts often face high costs, long implementation periods, and delayed ecological responses. Therefore, future restoration initiatives should prioritize the reconstruction of ecosystem structure and the reestablishment of key ecological functions while integrating natural succession dynamics along with socio-economic considerations [63]. Establishing green industries and ecological-cultural landscapes can enhance restoration outcomes and support local development, thereby creating a positive feedback cycle among ecological, economic, and social systems.

Restoration strategies should be differentiated according to disturbance type. In ongoing mining disturbance areas, such as core mining zones where NDVI remains persistently low, priority should be given to soil amelioration and substrate reconstruction, combined with the establishment of stress-tolerant vegetation communities and hydrological regulation measures, to promote long-term ecosystem recovery. In postmining areas with improving NDVI, restoration should emphasize natural succession with minimal artificial intervention. Long-term monitoring and ecological compensation mechanisms can guide vegetation communities toward stable and resilient states [64]. On the contrary, in construction-related disturbance areas, they generally exhibit short-term NDVI declines. Management should focus on strict spatial control, the establishment of ecological buffer zones, and the incorporation of green infrastructure to maintain landscape connectivity and reduce human interference [65].

From an international perspective, the postmining ecological restoration framework proposed by the Society for Ecological Restoration in 2022 provides a valuable reference for achieving socially and environmentally responsible restoration outcomes [66]. Future work may build on this framework to develop tiered restoration strategies tailored to specific disturbance types, thereby improving restoration efficiency,

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long-term stability, and overall ecological resilience in mining areas.

C. Limitations

This study employed the GEE platform and Landsat imagery, applying an improved ITSLT algorithm to construct dense NDVI time series trajectories for identifying and monitoring mining disturbances. The results demonstrate that NDVI-based trajectories effectively capture vegetation degradation and recovery processes, showing high sensitivity to mining-induced changes and postreclamation dynamics.

Compared with traditional LandTrendr approaches based on sparse annual observations, the proposed ITSLT framework improves temporal continuity and enhances the detection of subtle and short-term disturbance signals. By incorporating dense time series information, the method enables earlier identification of disturbance onset and more detailed characterization of disturbance–recovery trajectories. In addition, compared with single-index approaches, the trajectory-based analysis framework provides a more process-oriented understanding of ecological change, which is consistent with recent advances in time-series-based disturbance monitoring studies. The results of this study are in line with these findings, further demonstrating the advantages of integrating dense time-series analysis with trajectory classification for improving disturbance detection accuracy and ecological interpretation [8], [67].

Despite these strengths, several limitations remain. First, NDVI primarily reflects surface vegetation conditions and may have limited sensitivity to disturbances with weak or indirect surface expression, such as underground mining activities. In addition, uncertainties may arise from data-related factors, including residual cloud contamination after masking, and seasonal variability that may affect the representativeness of annual composites. Although the use of median compositing reduces noise, it may also smooth short-term extreme events. Second, NDVI-based analysis is subject to inherent constraints in representing vegetation conditions across different environments. For example, in areas with dense vegetation cover, the sensitivity of NDVI to further increases in biomass may be reduced, while in sparsely vegetated regions, background signals from soil or nonvegetated surfaces may influence the stability of the time series. Third, the current framework is based on pixel-level trajectory analysis, which may still be affected by spatial noise such as the "salt-and-pepper" effect. In addition, the parameterization of the ITSLT model and threshold selection may influence the detection results, particularly in areas with complex or overlapping disturbance processes.

To address these limitations, future research may pursue the following directions.

1) *Data integration:* combine Landsat with MODIS imagery through spatio-temporal fusion algorithms to enhance the temporal continuity and accuracy of NDVI trajectories.
2) *Methodological enhancement:* incorporate object-based change detection approaches to mitigate salt-and-pepper noise in pixel-level analyses.
3) *Indicator refinement:* explore alternative or complementary vegetation indices (e.g., enhanced vegetation index, vegetation index), incorporate environmental parameters (e.g., surface temperature, soil erosion), or integrate other time-series algorithms (e.g., CCDC, vegetation change tracker, detecting breakpoints and estimating segments in trend) to improve disturbance detection accuracy from a spatio-temporally continuous perspective.

Implementing these improvements would further enhance the stability and precision of surface-change monitoring and provide more reliable technical support for identifying mining-induced disturbances and evaluating restoration outcomes in the context of national land management and spatial planning.

V. Conclusion

This study developed a remote sensing-based monitoring framework for identifying and assessing ecological disturbances and recovery processes in mining areas, using an improved time-series analysis approach. By optimizing the LandTrendr algorithm, the proposed ITSLT model constructs dense NDVI time series from the full Landsat archive and dynamically integrates disturbance detection results, thereby enhancing sensitivity to subtle ecological changes and improving the spatio-temporal accuracy of disturbance identification. The model effectively captures weak disturbance signals that are often missed by traditional methods and accurately detects key change points at early disturbance stages, reducing both omission and commission errors.

Using this framework, trajectory-based classification was further applied to distinguish different disturbance types within the study area. The results reveal pronounced spatial heterogeneity in disturbance magnitude and recovery trajectories. Human-induced disturbances account for 52.91% of all events while mining-related disturbances contribute 2.61% and 5.98% of the total area, highlighting the dominant role of anthropogenic activities. Although vegetation conditions show an overall recovery trend, they have not yet returned to the levels observed in the early 2000's. Meanwhile, disturbance patterns have shifted from concentration in core mining zones to expansion along transportation and construction corridors, indicating a transition from localized disturbances to more spatially connected anthropogenic impacts.

From an application perspective, the proposed framework provides a clear technical pathway for dynamic ecological monitoring and management in mining areas. Dense time-series analysis enables early detection of disturbance signals in active mining zones, supporting timely intervention and regulation. In postmining areas, the framework can be used to evaluate restoration effectiveness and identify priority regions for ecological rehabilitation. In addition, integrating disturbance information with land-use expansion patterns offers practical support for coordinated planning of mining activities, infrastructure development, and ecological protection.

Overall, the integration of ITSLT and DTW offers a novel technical pathway for identifying mining-induced disturbances with improved spatio-temporal precision. This approach enhances the reliability of ecological change detection and pro-

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vides strong technical support and decision-making guidance for ecological restoration monitoring, ecological compensation assessment, and refined management of mining areas.

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